1. (a) Evaluate the indefinite integral

$$\int \frac{e^{\sqrt{x}}\cos(\sqrt{x})}{\sqrt{x}}\,dx.$$

[7 marks]

(b) Using your answer from part (a), show that

$$\int_0^{\pi^2} \frac{e^{\sqrt{x}}\cos(\sqrt{x})}{\sqrt{x}}\,dx = -(1+e^{\pi}).$$

What is the geometric interpretation of the value of the integral in part (b)?

[3 marks]

[Total 10 marks]

Your Solution:

$u=\sqrt{x}$

$du = \frac{1}{2\sqrt{x}}dx = \int \frac{e^u \cos u}{\cancel{u}} \cdot 2u\,du \rightarrow 2\int e^u \cos u\,du$

LIATE

$2\sqrt{x}\,du = dx$

$2u\,du = dx$

$w = \cos u \qquad v = e^u$

$dw = -\sin u\,du \qquad dv = e^u$

$\int e^u \cos u\,du = \cos u\, e^u + \int e^u \sin u\,du$

$w = \sin u \qquad v = e^u$

$dw = \cos u\,du \qquad dv = e^u$

$\int e^u \cos u\,du = e^u \cos u + \left[e^u \sin u - \int e^u \cos u\,du\right]$

$\int e^u \cos u\,du = e^u \cos u + e^u \sin u - \int e^u \cos u\,du$

$2\int e^u \cos u\,du = \frac{2e^u \cos u + e^u \sin u}{2} + C$

$$\boxed{\int \frac{e^{\sqrt{x}}\cos\sqrt{x}}{\sqrt{x}}\,dx = \frac{e^{\sqrt{x}}\cos\sqrt{x} + e^{\sqrt{x}}\sin\sqrt{x}}{2} + C}$$

Continue your solution to question 1 here...

b) $e^{\sqrt{x}}\cos\sqrt{x} + e^{\sqrt{x}}\sin\sqrt{x} \Big|_0^{\pi^2}$

$= \left[e^{\sqrt{\pi^2}}\left(\cos\sqrt{\pi^2} + \sin\sqrt{\pi^2}\right)\right] - \left[e^0\left(\cos 0 + \sin 0\right)\right]$

$= e^{\pi}(-1+0) - 1(1+0)$

$= -e^{\pi} - 1$

$= -(1+e^{\pi})$ ✓

what is a geometric interpretation?

triangular idk ?

Q2 10

2. Evaluate the following integrals.

(a) $\int \tan^4 x \, dx$. [7 marks]

(b) $\int_0^{\pi/4} \tan^4 x \, dx$. [3 marks]

[Total 10 marks]

Your Solution:

a) $\int \tan^4 x\,dx \to \int (\sec^2 x - 1)^2 dx \to \int (\sec^4 x - 2\sec^2 x + 1)\,dx$

$= \int \sec^4 x\,dx - 2\int \sec^2 x\,dx + \int dx$

$= \int (\tan^2 x + 1)\sec^2 x\,dx$ $\quad -2\int \sec^2 x\,dx = -2\tan x$ $\quad = x$

$u = \tan x \quad du = \sec^2 x\,dx$

$= \int (u^2+1)\,du$

$= \frac{u^3}{3} + u$

$= \frac{\tan^3 x}{3} + \tan x$

$= \frac{\tan^3 x}{3} + \tan x - 2\tan x + x + C$

$= \boxed{\frac{\tan^3 x}{3} - \tan x + x + C}$ ✓

Continue your solution to question 2 here…

b) $\int_0^{\pi/4} \tan^4 x\,dx = \frac{\tan^3 x}{3} - \tan x + x \Big|_0^{\pi/4}$

$= \left[\frac{(\tan\frac{\pi}{4})^3}{3} - \tan\frac{\pi}{4} + \frac{\pi}{4}\right] - \left[\frac{(\tan 0)^3}{3} - \tan(0) + 0\right]$

$= \frac{1}{3} - 1 + \frac{\pi}{4}$

$= \boxed{-\frac{2}{3} + \frac{\pi}{4}}$

Q3 10

3. (a) Evaluate

$$\int \frac{x}{(x-3)^3}\,dx.$$

[4 marks]

(b) Explain why the following definite integral is an improper integral.

$$I = \int_1^3 \frac{x}{(x-3)^3}\,dx.$$

[2 marks]

(c) Show whether I is convergent or divergent. [4 marks]

[Total 10 marks]

Your Solution:

$$\int \frac{x}{(x-3)^3}dx = \frac{A}{x-3} + \frac{B}{(x-3)^2} + \frac{C}{(x-3)^3}$$

$$x = A(x-3)^2 + B(x-3) + C$$

$$x = A(x^2-6x+9) + Bx - 3B + C$$

$$x = Ax^2 - 6Ax + 9A + Bx - 3B + C$$

$$x = (A)x^2 + (-6A+B)x + (9A-3B+C)$$

$A=0$

$-6A + B = 1 \rightarrow B = 1$

$9(0) - 3(1) + C = 0 \rightarrow C = 3$

$A=0$
$B=1$
$C=3$

Continue your solution to question 3 here...

$$\rightarrow \int \frac{x}{(x-3)^3}dx = \int \frac{1}{(x-3)^2} + \frac{3}{(x-3)^3}\,dx$$

$$= \int \frac{1}{(x-3)^2}dx + 3\int \frac{1}{(x-3)^3}dx$$

b) it is improper because it is discontinuous at $x=3$ which is one of the bounds

$u = x-3$
$du = dx$

$$= \int \frac{1}{u^2}du$$

$$= \int u^{-2}du$$

$$= -\frac{1}{(x-3)}$$

$v = x-3$
$dv = dx$

$$= 3\int \frac{1}{v^3}dv$$

$$= 3\int v^{-3}dv$$

$$= 3 \cdot \frac{v^{-2}}{-2} = -\frac{3}{2(x-3)^2}$$

$$= \boxed{-\frac{1}{x-3} - \frac{3}{2(x-3)^2} + C}$$

c) $$= \lim_{c\to 3}\left[-\frac{1}{x-3} - \frac{3}{2(x-3)^2}\right]_1^c$$

$$= \lim_{c\to 3}\left[-\frac{1}{c-3} - \frac{3}{2(c-3)^2}\right] - \left[-\frac{1}{1-3} - \frac{3}{2(1-3)^2}\right]$$

$$= \lim_{c\to 3}\left[\frac{-2(c-3)^2 - 3(c-3)}{2(c-3)^3}\right] - \left[-\frac{1}{-2} - \frac{3}{2(-2)^2}\right]$$

$$= \lim_{c\to 3}\left[\frac{-2(c-3) - 3}{2(c-3)^3}\right] - \left[\frac{1}{2} - \frac{3}{8}\right]$$

not indeterminate
$\therefore$ cannot L'hopital

$\therefore \lim_{c\to 3}$ DNE, $\int \frac{x}{(x-3)^3}dx$ DIVERGENT

Q4 8

4. Consider the integral

$$I = \int \frac{x^2}{\sqrt{16 - x^2}} dx.$$

(a) What integration technique could be used to evaluate the integral I.

[2 marks]

(b) Evaluate the integral I. [6 marks]

[Total 8 marks]

Your Solution:

a) the trigonometric substitution technique can be used where $x = 4\sin\theta$ to solve the integral

b) $x = 4\sin\theta$ $\rightarrow$ $= \int \frac{16\sin^2\theta}{\sqrt{16 - 16\sin^2\theta}} 4\cos\theta d\theta$

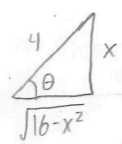
$dx = 4\cos\theta d\theta$

$\sin = \frac{o}{h}$

$\sin\theta = \frac{x}{4}$

$= \int \frac{16\sin^2\theta}{\cancel{4\cos\theta}} \cdot \cancel{4\cos\theta} d\theta = 16\int \sin^2\theta d\theta = -16\int \frac{1}{2}(1 - \cos 2\theta) d\theta$

$= -8\int (1 - \cos 2\theta) d\theta$



$= -8\int 1 d\theta + 8\int \cos 2\theta d\theta$

$= -8\theta + 8\frac{\sin 2\theta}{2}$

$= -8\theta + 4\sin 2\theta$

$= -8\theta + 4 \cdot 2\sin\theta\cos\theta$

$= -8(\theta - \sin\theta\cos\theta)$

$= \boxed{-8\left[\sin^{-1}\left(\frac{x}{4}\right) - \left(\frac{x}{4}\right)\left(\frac{\sqrt{16-x^2}}{4}\right)\right] + C}$

Q5 7

5. Evaluate the integral

$$\int \frac{2x+1}{x^2+2x-8}dx. \rightarrow \int \frac{2x+1}{(x+4)(x-2)}dx$$

[Total 7 marks]

Your Solution:

$$\int \frac{2x+1}{(x+4)(x-2)}dx = \int \frac{A}{x+4} + \frac{B}{x-2}dx$$

$$2x+1 = A(x-2) + B(x+4)$$

Test $x=2$: $4+1 = A(2-2) + B(2+4) \rightarrow 5 = 6B$ $\boxed{B = \frac{5}{6}}$

Test $x=-4$ $2(-4)+1 = A(-4-2) + B(-4+4) \rightarrow -7 = -6A$ $\boxed{A = \frac{7}{6}}$

$$\int \frac{2x+1}{(x+4)(x-2)}dx = \int \frac{\frac{7}{6}}{x+4} + \frac{\frac{5}{6}}{x-2}dx \rightarrow \frac{7}{6}\int \frac{1}{x+4}dx + \frac{5}{6}\int \frac{1}{x-2}dx$$

$$= \frac{7\ln|x+4|}{6} + \frac{5\ln|x-2|}{6}$$

$$= \boxed{\frac{7\ln|x+4| + 5\ln|x-2|}{6} + C}$$

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Continue your solution to question 5 here…

Q6 5

6. Find an implicit general solution of the following differential equation.

$$\frac{du}{dt} = \frac{1+t^4}{ut^2+u^4t^2}.$$

[Total 5 marks]

Your Solution:

$$\frac{du}{dt} = \frac{1+t^4}{t^2(u+u^4)} \rightarrow (u+u^4)du = \left(\frac{1+t^4}{t^2}\right)dt$$

$$\int(u+u^4)du = \int \frac{1+t^4}{t^2}dt \rightarrow \int(u+u^4)du = \int\left(\frac{1}{t^2}+t^2\right)dt$$

$$\frac{u^2}{2}+\frac{u^5}{5} = \int\frac{1}{t^2}dt + \int t^2 dt \rightarrow \frac{u^2}{2}+\frac{u^5}{5} = -\frac{1}{t}+\frac{t^3}{3}+C$$

$$\boxed{\frac{5u^2+2u^5}{10} = \frac{t^3}{3}-\frac{1}{t}+C}$$

Q7 10

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7. Consider the following differential equation,

$$x^2y' + 2xy = \ln x\,.$$

(a) Classify the differential equation as either *directly integrable differential equation*, *separable differential equation*, or *first-order linear differential equation*. Justify your answer. [2 marks]

(b) Solve the initial value problem,

$$x^2y' + 2xy = \ln x\,, \text{ where } y(1) = 2.$$

[8 marks]

[Total 10 marks]

Your Solution:

a) it is a first order LDE because it follows the format of

$$\frac{dy}{dx} + P(x)y = Q(x)$$

b) $x^2\frac{dy}{dx} + 2x \cdot y = \ln x \rightarrow \frac{dy}{dx} + \frac{2x}{x^2}y = \frac{\ln x}{x^2}$

$\rightarrow \frac{dy}{dx} + \frac{2}{x}y = \frac{\ln x}{x^2}$ $\quad I = e^{\int \frac{2}{x}dx} = e^{2\int \frac{1}{x}dx} = e^{2\ln|x|} = e^{\ln|x|^2} = x^2$

$y(x) = \frac{1}{x^2}\left[\int x^2 \cdot \frac{\ln x}{x^2}dx + C\right] = \frac{1}{x^2}\left[\int \ln x\,dx + C\right] = \frac{1}{x^2}\left[(x\ln x - x) + C\right.$

$y(x) = \frac{x(\ln x - 1)}{x^2} + \frac{C}{x^2} = \frac{\ln x - 1}{x} + \frac{C}{x^2}$

$2 = \frac{\ln(1)^0 - 1}{1} + \frac{C}{1} \rightarrow 2 = -1 + C \quad C = 3$

$$\boxed{y(x) = \frac{\ln x - 1}{x} + \frac{3}{x^2}}$$

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